

Scope — taxonomy-collapse re-run

Results

Measured, not estimated: the collapse lowers the headline score. Folding given_name / family_name / middle_name into full_name and street_number_and_name into address, in both gold and predictions, and dropping the four folded tags from each catalogue:

Arm	macro F2	→ collapsed	micro F1	→ collapsed	Gates
champion_1k	0.4835	0.4745 (−0.0090)	0.3812	0.3642 (−0.0170)	55/55 → 55/55
perlabel_v4	0.4932	0.4855 (−0.0077)	0.4249	0.4123 (−0.0126)	55/55 → 55/55

This was re-scored from cached predictions under the collapsed mapping. **No evaluator was changed and no model was selected** — it is a probe, run so the payoff could be known before committing.

The mechanism works exactly as predicted — and still loses

The merge does what the agreement analysis said it would. On the corpora where annotators used given_name without full_name, a full_name prediction was being scored as a false positive; after the fold it is a true positive:

Corpus	Tag	Support	Precision
pii_holdout_20k	full_name	7,836 → 11,449	0.388 → 0.571
pii2_eval_30k	full_name	6,811 → 9,994	0.229 → 0.339
pii_holdout_20k	address	4,629 → 6,566	0.232 → 0.329
openpii_pii_eval_38k	full_name	29,335 → 30,464	0.753 → 0.782
ai4privacy	full_name	5,114 → 5,470	0.483 → 0.516
pii2_eval_30k	address	2,553 → 2,841	0.085 → 0.094

full_name precision rises **47%** on pii_holdout_20k and **48%** on pii2_eval_30k. Recall is unchanged throughout. That is a real measurement error being corrected on the highest-support priority tag in the project.

It loses anyway, because of the denominator. macro_f2 averages over the frozen corpus catalogue, so removing four tags changes the mean regardless of what else happens — and those four tags are *not* weak:

Corpus	family	given	middle	street	Effect of dropping them
openpii	0.908	0.928	0.889	0.866	−0.0344
pii_holdout_20k	0.870	0.889	0.830	0.845	−0.0219
pii2	0.786	0.747	0.701	0.721	−0.0178
ai4privacy	0.468	0.534	0.096	0.656	+0.0141

Corpus	family	given	middle	street	Effect of dropping them
betterdataai	0.071	0.171	0.000	0.426	+0.0136
					net -0.0093

On three of the five complete corpora the model scores **0.70–0.93** on the name components *separately*. The collapse throws that credit away, and the merge gain on two tags does not cover the loss of four.

So the finding is about what you want to measure

The agreement probe and this probe are both right, and they are not measuring the same thing:

- **Agreement** rises under the collapse (macro 0.514 → 0.635, weighted 0.469 → 0.788) because two *judges* cannot agree on which name tag to use.
- **The model's score** falls, because the model *can* reproduce the distinction — it was trained on these corpora and has learned each one's convention.

That is the uncomfortable part worth stating plainly: the model scoring 0.89 on `given_name` where two independent judges agree at 0.29 does not mean the model knows something the judges do not. It means it has learned **this corpus's labelling convention**, which is exactly the thing that does not generalise. Collapsing removes the reward for convention-fitting — and the score falls because a good part of the score was convention-fitting.

A lower number that measures a concept the labels can pin down is worth more than a higher one that rewards reproducing an arbitrary convention. But that is a judgement about the contract, not something a probe settles, and it is yours to make.

TL;DR

- **The collapse lowers macro F2 by 0.008 and micro F1 by 0.013.** Measured on cached predictions for both arms; gates stay 55/55 either way.
- The intended mechanism works — `full_name` precision **+47%** on `pii_holdout_20k`, **+48%** on `pii2_eval_30k`, recall unchanged — but is outweighed by dropping four tags the model scores 0.70–0.93 on.
- **The earlier "ceiling rises to 0.635" claim was about judge agreement, not about the model's score.** Both are true; they are different quantities. The score falls precisely because part of it was reproducing an arbitrary labelling convention that two independent judges could not agree on.
- **Compute is not the constraint** — all 12 arms have cached predictions, so a full re-run is ~20 minutes. The cost is that every published number becomes non-comparable, including the eight `verify.py` pins and the shipped model card.
- **Recommendation: option C.** Keep the contract, report the collapsed view beside it. If the goal is a number that generalises rather than a number that is high, option B is defensible — but it should be chosen for that reason, not in the expectation that the score goes up.

Model scoreboard

Model	Dataset	F1@5	F1@3	F1@1	Docs/sec	ms/doc
current_rules	4000_datax-dualjudge-evalset-1.32k					
current_rules	6589_govdocs2-dualjudge-eval20-3.53k					
current_rules	ai4privacy_pii_masking_eval_10k					
current_rules	betterdataai_ner_silver_eval_10k					
current_rules	nemotron_eval_6k					
current_rules	openpii_pii_eval_38k					
current_rules	pii2_eval_30k					
current_rules	pii_holdout_20k					
hash_sgd	4000_datax-dualjudge-evalset-1.32k					
hash_sgd	6589_govdocs2-dualjudge-eval20-3.53k					
hash_sgd	ai4privacy_pii_masking_eval_10k					
hash_sgd	betterdataai_ner_silver_eval_10k					
hash_sgd	nemotron_eval_6k					
hash_sgd	openpii_pii_eval_38k					
hash_sgd	pii2_eval_30k					
hash_sgd	pii_holdout_20k					
hash_sgd_f2	4000_datax-dualjudge-evalset-1.32k					
hash_sgd_f2	6589_govdocs2-dualjudge-eval20-3.53k					
hash_sgd_f2	ai4privacy_pii_masking_eval_10k					
hash_sgd_f2	betterdataai_ner_silver_eval_10k					
hash_sgd_f2	nemotron_eval_6k					
hash_sgd_f2	openpii_pii_eval_38k					
hash_sgd_f2	pii2_eval_30k					
hash_sgd_f2	pii_holdout_20k					
hybrid_priority_001	4000_datax-dualjudge-evalset-1.32k					
hybrid_priority_001	6589_govdocs2-dualjudge-eval20-3.53k					
hybrid_priority_001	ai4privacy_pii_masking_eval_10k					
hybrid_priority_001	betterdataai_ner_silver_eval_10k					
hybrid_priority_001	nemotron_eval_6k					

Model	Dataset	F1@5	F1@3	F1@1	Docs/sec	ms/doc
hybrid_priority_001	openpii_pii_eval_38k					
hybrid_priority_001	pii2_eval_30k					
hybrid_priority_001	pii_holdout_20k					
tfidf_linear	4000_datax-dualjudge-evalset-1.32k					
tfidf_linear	6589_govdocs2-dualjudge-eval20-3.53k					
tfidf_linear	ai4privacy_pii_masking_eval_10k					
tfidf_linear	betterdataai_ner_silver_eval_10k					
tfidf_linear	nemotron_eval_6k					
tfidf_linear	openpii_pii_eval_38k					
tfidf_linear	pii2_eval_30k					
tfidf_linear	pii_holdout_20k					
embeddingbag_asl	4000_datax-dualjudge-evalset-1.32k					
embeddingbag_asl	6589_govdocs2-dualjudge-eval20-3.53k					
embeddingbag_asl	ai4privacy_pii_masking_eval_10k					
embeddingbag_asl	betterdataai_ner_silver_eval_10k					
embeddingbag_asl	nemotron_eval_6k					
embeddingbag_asl	openpii_pii_eval_38k					
embeddingbag_asl	pii2_eval_30k					
embeddingbag_asl	pii_holdout_20k					
hybrid_priority	4000_datax-dualjudge-evalset-1.32k					
hybrid_priority	6589_govdocs2-dualjudge-eval20-3.53k					
hybrid_priority	ai4privacy_pii_masking_eval_10k					
hybrid_priority	betterdataai_ner_silver_eval_10k					
hybrid_priority	nemotron_eval_6k					
hybrid_priority	openpii_pii_eval_38k					
hybrid_priority	pii2_eval_30k					
hybrid_priority	pii_holdout_20k					
champion_1k	4000_datax-dualjudge-evalset-1.32k					
champion_1k	6589_govdocs2-dualjudge-eval20-3.53k					

Model	Dataset	F1@ 5	F1@ 3	F1@ 1	Docs/s ec	ms/ doc
champion_1k	ai4privacy_pii_masking_eval_10k					
champion_1k	betterdataai_ner_silver_eval_10k					
champion_1k	nemotron_eval_6k					
champion_1k	openpii_pii_eval_38k					
champion_1k	pii2_eval_30k					
champion_1k	pii_holdout_20k					
champion_1k_bootstrap	8 holdouts (5 complete for F2)				454.5	2.2
hash_sgd_readpass	8 holdouts / fixed 1,000-doc latency sample				984.8	1.0
hash_sgd_readpass	8 holdouts / fixed 1,000-doc latency sample				483.5	2.1
hash_sgd_readpass	8 holdouts / fixed 1,000-doc latency sample				204.6	4.9
hash_sgd_readpass	8 holdouts / fixed 1,000-doc latency sample				146.2	6.8
hybrid_priority_001_readpass	8 holdouts / fixed 1,000-doc latency sample				674.0	1.5
hybrid_priority_001_readpass	8 holdouts / fixed 1,000-doc latency sample				341.6	2.9
hybrid_priority_001_readpass	8 holdouts / fixed 1,000-doc latency sample				175.4	5.7
hybrid_priority_001_readpass	8 holdouts / fixed 1,000-doc latency sample				130.7	7.6
hybrid_priority_readpass	8 holdouts / fixed 1,000-doc latency sample				454.5	2.2
hybrid_priority_readpass	8 holdouts / fixed 1,000-doc latency sample				237.5	4.2
hybrid_priority_readpass	8 holdouts / fixed 1,000-doc latency sample				147.8	6.8
hybrid_priority_readpass	8 holdouts / fixed 1,000-doc latency sample				113.3	8.8
perlabel_v4	4000_datax-dualjudge-evalset-1.32k					
perlabel_v4	6589_govdocs2-dualjudge-eval20-3.53k					
perlabel_v4	ai4privacy_pii_masking_eval_10k					
perlabel_v4	betterdataai_ner_silver_eval_10k					
perlabel_v4	nemotron_eval_6k					
perlabel_v4	openpii_pii_eval_38k					
perlabel_v4	pii2_eval_30k					
perlabel_v4	pii_holdout_20k					

Measured 2026-08-25 · n=1,000, n=126,129.